# Team 02 – ResuMind AI

Resume Processing Pipeline: Converting unstructured PDF documents into structured JSON data using AI and intelligent text processing

# Meet Our Team

ResuMind AI brings together six specialized modules, each designed to handle a critical stage in the resume processing pipeline. From initial file upload to final validation, every team member ensures data quality and accuracy.

### Sayali

User Upload

### Akanksha

File Validation

### Pratik

PDF Text Extraction

### Divya

Text Normalization

### Vikrant

LLM Processing

### Umaima

Schema Validation


# Problem Statement

## The Challenge

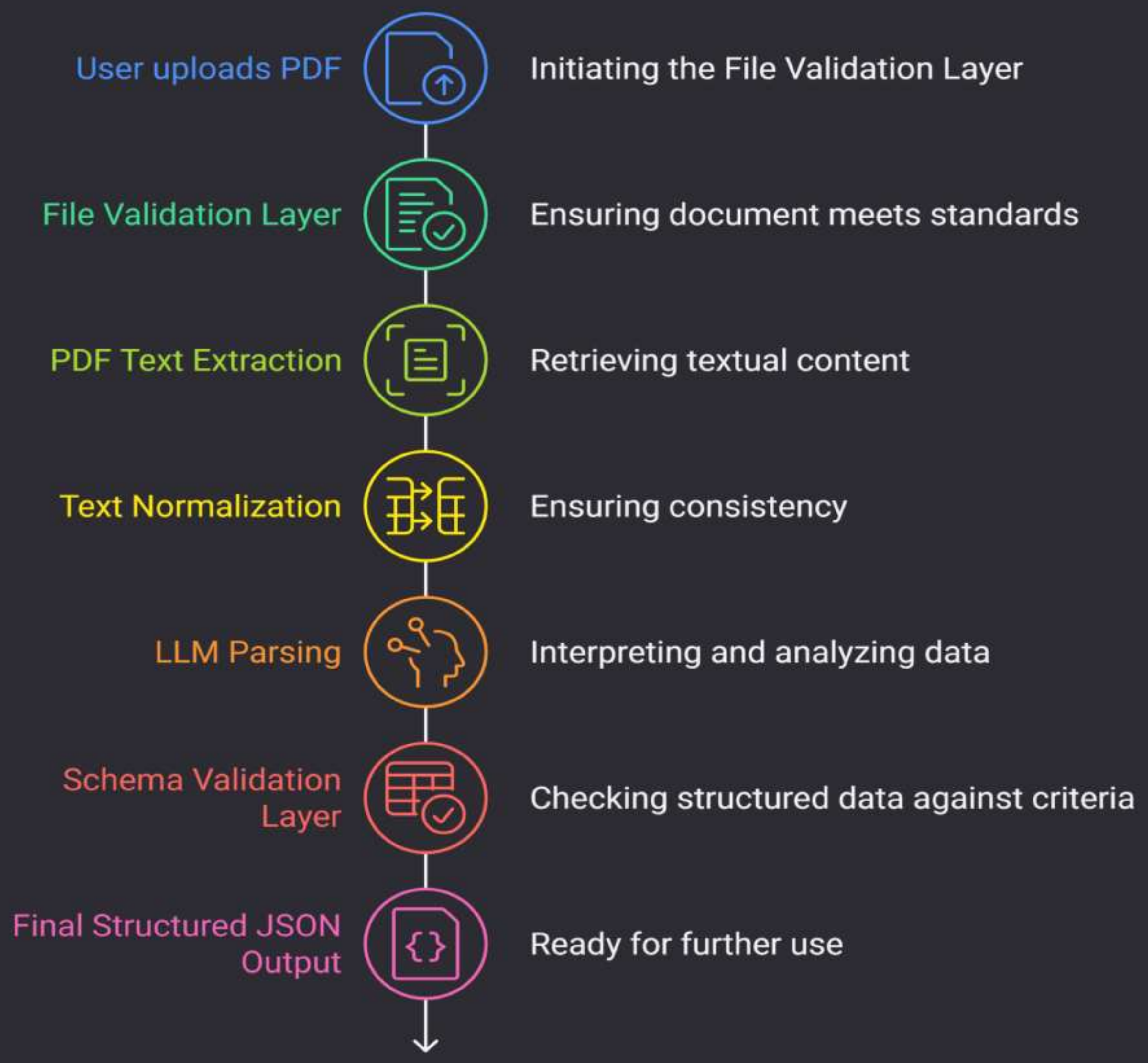
Resume data arrives in PDF format—unstructured, inconsistent, and impossible for AI systems to process directly. Each document follows different layouts, contains unique formatting quirks, and presents information in unpredictable ways.

## The Solution

Our pipeline transforms raw PDF documents into clean, structured JSON format that AI systems can understand and utilize. This conversion enables automated resume screening, skills matching, and candidate evaluation at scale.

1. **Unstructured PDF**
   Raw document with mixed formatting
2. **Text Extraction**
   Convert to readable format
3. **Cleaning & Normalization**
   Standardize structure
4. **Structured JSON**
   AI-ready data format

# Architecture




# Sayali – User Upload Module

## Core Responsibilities

Manages the initial file reception layer, ensuring reliable data transfer from end users to the processing pipeline.

## Key Features

- Accepts PDF and DOCX file formats
- Converts files into binary streams for processing
- Implements robust error handling for interrupted uploads
- Optimized stream processing for large files
- Provides immediate feedback on upload status

# Akanksha – File Validation

Acts as the first quality gate, preventing malformed or malicious files from entering the processing pipeline. This module performs comprehensive security and integrity checks before any processing begins.

### MIME Type Validation

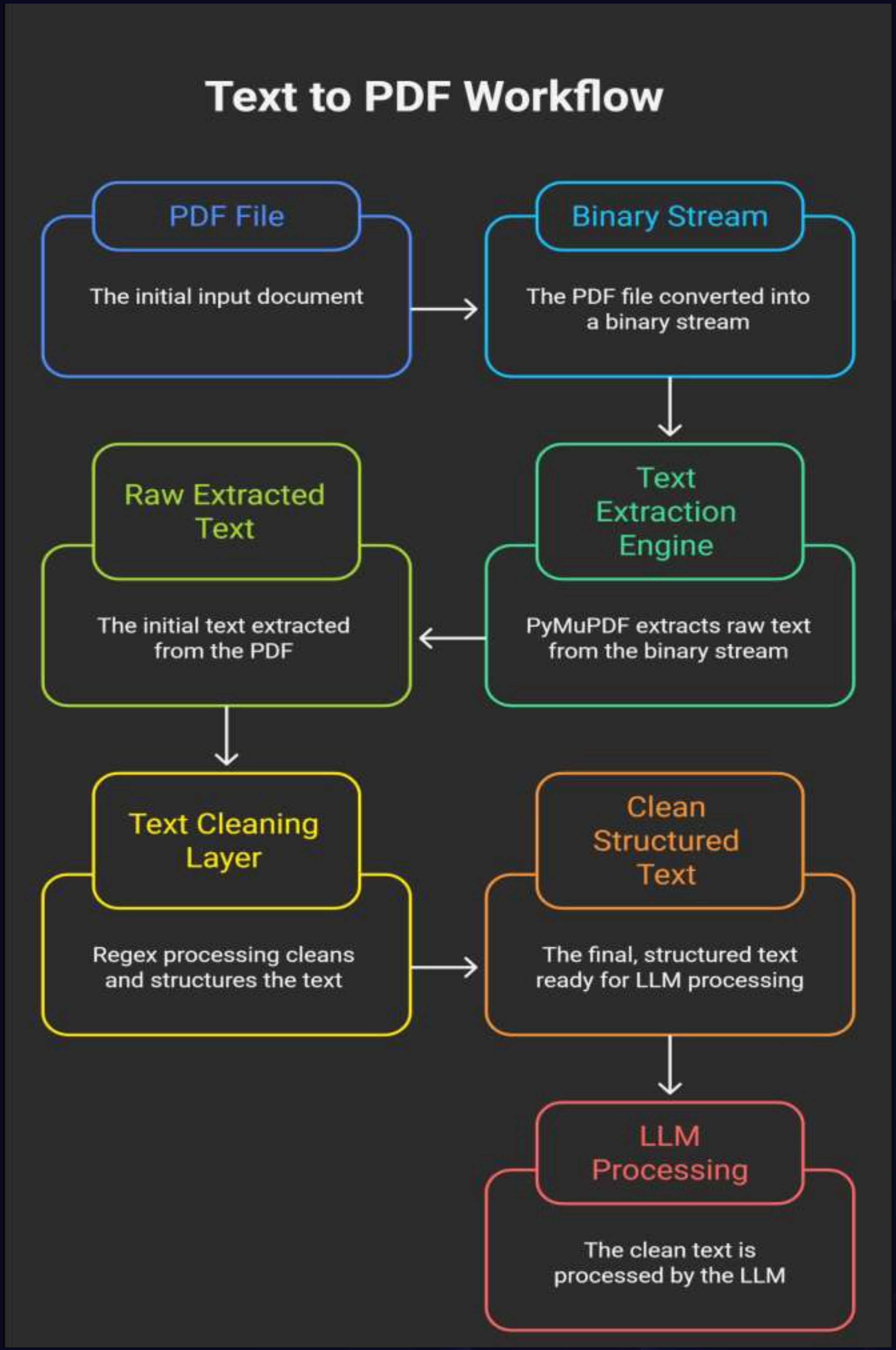
Verifies file type matches expected format, blocking potentially dangerous file extensions and ensuring only approved document types proceed to extraction.

### Size & Token Limits

Applies configurable constraints on file size and estimated token count, preventing resource exhaustion and ensuring processing efficiency across the pipeline.

### Corruption Detection

Identifies corrupt, empty, or password-protected files that cannot be processed, providing clear error messages to guide users toward valid resume formats.




# Pratik – PDF Text Extraction

Transforms binary PDF documents into readable text format, handling the complexities of multi-page layouts, column arrangements, and mixed content types that characterize modern resumes.

### PDF to Text Conversion

Utilizes PyMuPDF and PDFMiner libraries to accurately extract text content from binary PDF format

### Multi-Page Support

Handles multi-page resumes with consistent formatting and logical page sequencing

### Multi-Column Processing

Manages two-column and three-column resume layouts, preserving reading order and structure

### Text Cleaning

Removes extra spaces ,tabs , line breaks and unwanted characters.

# Divya – Text Normalization Module

## Cleaning Process

Raw text extracted from PDFs contains numerous formatting artifacts that interfere with downstream processing. This module systematically removes these inconsistencies while preserving semantic meaning.

- Eliminates extra whitespace and line breaks
- Removes special characters and symbols
- Standardizes text to lowercase
- Fixes encoding issues and character sets
- Normalizes date and number formats

## Regular Expression Engine

Employs sophisticated regex patterns to identify and remove unwanted patterns while preserving meaningful content structure.


# Vikrant – LLM Processing Module

Converts normalized text into structured JSON format using large language models with specialized prompts and constraints. This is where unstructured text becomes machine-readable data.

### Structured JSON Output

Generates well-formed JSON with consistent field names and data types

### Entity Extraction

Identifies and extracts names, skills, experience, education, and contact information

### Security Measures

Includes prompt injection safeguards and token limit handling

# Umaima – Schema Validation Module

The final quality gate ensures production-ready JSON output. This module verifies data structure, handles missing fields, and corrects formatting issues before the data enters downstream systems.

### Type Checking

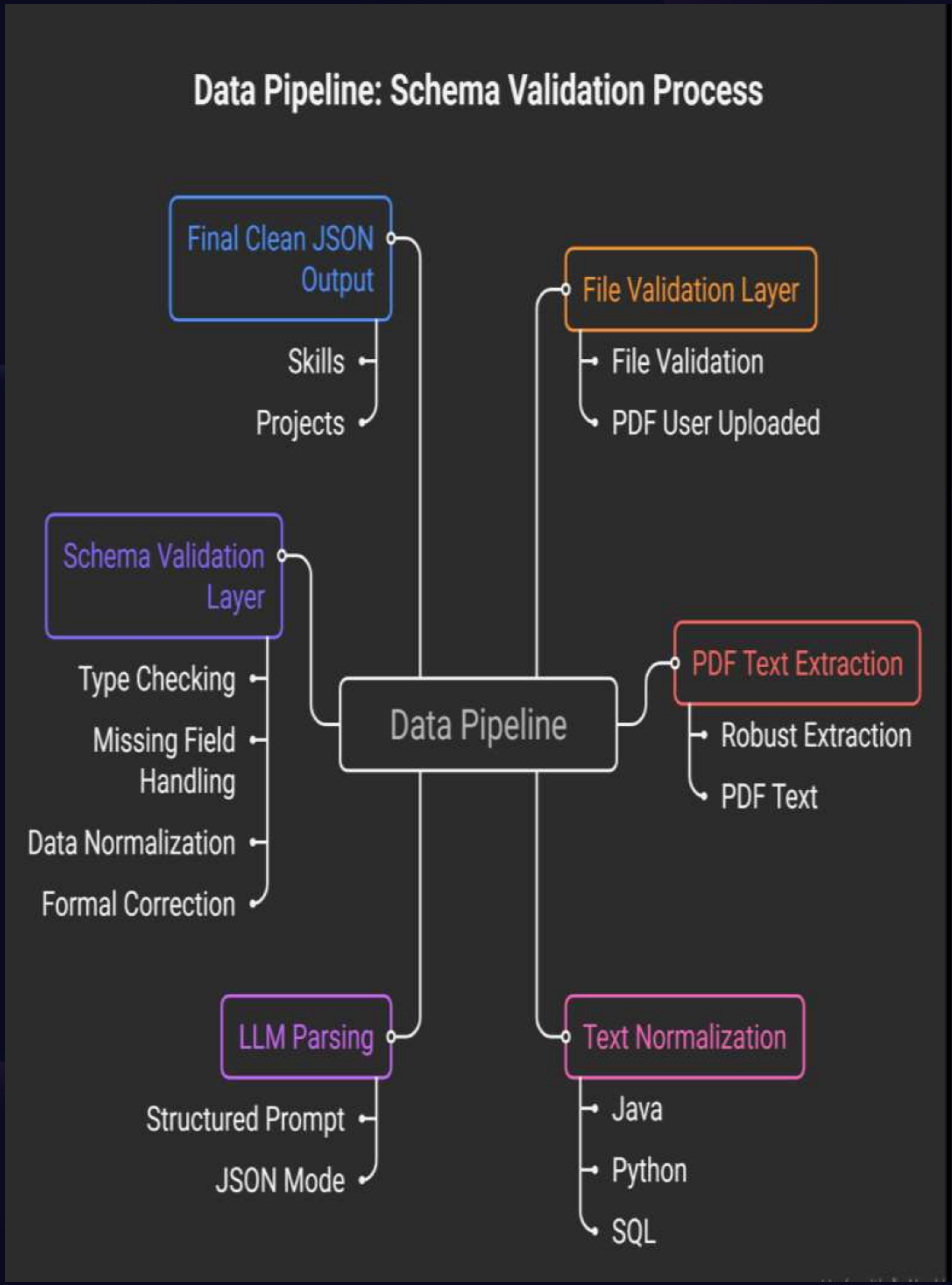
Validates that all fields match expected data types, ensuring skills are stored as lists, dates follow ISO format, and numerical values are properly typed

### Missing Field Handling

Implements intelligent fallback logic for optional fields, providing default values or null handling strategies based on business requirements

### Data Normalization

Standardizes values across consistent formats, ensuring skills use controlled vocabulary, locations follow geographic standards, and company names are properly formatted



Thank You !